THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **A Square and a Cube** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 49	$7 \times 7 = 49$
2	**(b)** 144	$12 \times 12 = 144$
3	**(b)** 13	$13 \times 13 = 169$
4	**(d)** 64	$4 \times 4 \times 4 = 64$
5	**(c)** 4	$4^3 = 64$
6	**(c)** 25	Sum of first n odd numbers $= n^2 = 25$
7	**(b)** 64	$8 \times 8 = 64$
8	**(b)** 196	$14 \times 14 = 196$
9	**(b)** 17	$17 \times 17 = 289$
10	**(b)** 125	$5 \times 5 \times 5 = 125$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$27^2 = 27 \times 27 = 729$ (2).	2
12	Yes, $225 = 15 \times 15$ (1+1).	1+1
13	$13^3 = 13 \times 13 \times 13 = 2197$ (2).	2
14	$33^2 = 33 \times 33 = 1089$ (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$196 = 2^2 \times 7^2$ (1); pairing equal factors (1); $\sqrt{196} = 14$ (1).	1+1+1
16	$512 = 2^9$ (1); grouping equal factors in triplets (1); $\sqrt[3]{512} = 8$ (1).	1+1+1
17	$1 + 3 + 5 + 7 + 9 + 11 = 36$ (3).	3
	OR: $324 = 2^2 \times 3^4$ (1); pairing equal factors (1); $\sqrt{324} = 18$ (1).	1+1+1
18	$729 = 3^6$ (1); grouping equal factors in triplets (1); $\sqrt[3]{729} = 9$ (1).	1+1+1
	OR: $1 + 3 + 5 + 7 + 9 + 11 + 13 = 49$ (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$180 = 2^2 \times 3^2 \times 5$ (1); primes with odd power: multiplier $= 5$ (2); product $= 900$ (1); square root $= 30$ (1).	1+2+1+1
	OR: $108 = 2^2 \times 3^3$ (1); smallest divisor $= 4$ (2); quotient $= 27$ (1); cube root $= 3$ (1).	1+2+1+1
20	$252 = 2^2 \times 3^2 \times 7$ (1); primes with odd power: multiplier $= 7$ (2); product $= 1764$ (1); square root $= 42$ (1).	1+2+1+1
	OR: $500 = 2^2 \times 5^3$ (1); smallest divisor $= 4$ (2); quotient $= 125$ (1); cube root $= 5$ (1).	1+2+1+1

Mistakes to avoid:

- Concluding that a number is a perfect square from its last digit alone: a units digit of 0, 1, 4, 5, 6 or 9 is only *necessary*, not sufficient.
- Pairing primes for a perfect cube: grouping them in **pairs** (square-root rule) instead of **triples**.
- Writing $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$ or $(a+b)^2 = a^2 + b^2$.
- Mixing up 2^3 with 2×3 or 3^2 with 3×2; $4^3 = 64$, not 12.
- Saying the cube root of -8 does not exist; it is -2 because $(-2)^3 = -8$.
- Multiplying or dividing by the wrong prime when making a number a perfect square or cube, instead of looking only at the primes with an incorrect exponent.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **A Square and a Cube** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 5	$5^3 = 125$
2	**(a)** 49	Sum of first n odd numbers $= n^2 = 49$
3	**(c)** 81	$9 \times 9 = 81$
4	**(d)** 256	$16 \times 16 = 256$
5	**(b)** 19	$19 \times 19 = 361$
6	**(a)** 343	$7 \times 7 \times 7 = 343$
7	**(b)** 6	$6^3 = 216$
8	**(c)** 81	Sum of first n odd numbers $= n^2 = 81$
9	**(a)** 100	$10 \times 10 = 100$
10	**(d)** 324	$18 \times 18 = 324$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	No (1), because it lies between $15^2 = 225$ and $16^2 = 256$ (1).	1+1
12	$15^3 = 15 \times 15 \times 15 = 3375$ (2).	2
13	$41^2 = 41 \times 41 = 1681$ (2).	2
14	Yes, $361 = 19 \times 19$ (1+1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$441 = 3^2 \times 7^2$ (1); pairing equal factors (1); $\sqrt{441} = 21$ (1).	1+1+1
16	$1728 = 2^6 \times 3^3$ (1); grouping equal factors in triplets (1); $\sqrt[3]{1728} = 12$ (1).	1+1+1
17	$1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 = 64$ (3).	3
	OR: $576 = 2^6 \times 3^2$ (1); pairing equal factors (1); $\sqrt{576} = 24$ (1).	1+1+1
18	$2197 = 13^3$ (1); grouping equal factors in triplets (1); $\sqrt[3]{2197} = 13$ (1).	1+1+1
	OR: $1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 = 81$ (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$300 = 2^2 \times 3 \times 5^2$ (1); primes with odd power: multiplier $= 3$ (2); product $= 900$ (1); square root $= 30$ (1).	1+2+1+1
	OR: $1080 = 2^3 \times 3^3 \times 5$ (1); smallest divisor $= 5$ (2); quotient $= 216$ (1); cube root $= 6$ (1).	1+2+1+1
20	$450 = 2 \times 3^2 \times 5^2$ (1); primes with odd power: multiplier $= 2$ (2); product $= 900$ (1); square root $= 30$ (1).	1+2+1+1
	OR: $1350 = 2 \times 3^3 \times 5^2$ (1); smallest divisor $= 50$ (2); quotient $= 27$ (1); cube root $= 3$ (1).	1+2+1+1

Mistakes to avoid:

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- Pairing primes for a perfect cube: grouping them in **pairs** (square-root rule) instead of **triples**.
- Writing $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$ or $(a+b)^2 = a^2 + b^2$.
- Mixing up 2^3 with 2×3 or 3^2 with 3×2; $4^3 = 64$, not 12.
- Saying the cube root of -8 does not exist; it is -2 because $(-2)^3 = -8$.
- Multiplying or dividing by the wrong prime when making a number a perfect square or cube, instead of looking only at the primes with an incorrect exponent.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **A Square and a Cube** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 22	$22 \times 22 = 484$
2	**(b)** 512	$8 \times 8 \times 8 = 512$
3	**(b)** 7	$7^3 = 343$
4	**(b)** 100	Sum of first n odd numbers $= n^2 = 100$
5	**(a)** 121	$11 \times 11 = 121$
6	**(c)** 441	$21 \times 21 = 441$
7	**(c)** 24	$24 \times 24 = 576$
8	**(c)** 1000	$10 \times 10 \times 10 = 1000$
9	**(d)** 8	$8^3 = 512$
10	**(a)** 144	$12 \times 12 = 144$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$16^3 = 16 \times 16 \times 16 = 4096$ (2).	2
12	$52^2 = 52 \times 52 = 2704$ (2).	2
13	No (1), because it lies between $22^2 = 484$ and $23^2 = 529$ (1).	1+1
14	$20^3 = 20 \times 20 \times 20 = 8000$ (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$784 = 2^4 \times 7^2$ (1); pairing equal factors (1); $\sqrt{784} = 28$ (1).	1+1+1
16	$2744 = 2^3 \times 7^3$ (1); grouping equal factors in triplets (1); $\sqrt[3]{2744} = 14$ (1).	1+1+1
17	$1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 + 21 = 121$ (3).	3
	OR: $900 = 2^2 \times 3^2 \times 5^2$ (1); pairing equal factors (1); $\sqrt{900} = 30$ (1).	1+1+1
18	$3375 = 3^3 \times 5^3$ (1); grouping equal factors in triplets (1); $\sqrt[3]{3375} = 15$ (1).	1+1+1
	OR: $1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 + 21 + 23 = 144$ (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$490 = 2 \times 5 \times 7^2$ (1); primes with odd power: multiplier $= 10$ (2); product $= 4900$ (1); square root $= 70$ (1).	1+2+1+1
	OR: $2000 = 2^4 \times 5^3$ (1); smallest divisor $= 2$ (2); quotient $= 1000$ (1); cube root $= 10$ (1).	1+2+1+1
20	$588 = 2^2 \times 3 \times 7^2$ (1); primes with odd power: multiplier $= 3$ (2); product $= 1764$ (1); square root $= 42$ (1).	1+2+1+1
	OR: $3240 = 2^3 \times 3^4 \times 5$ (1); smallest divisor $= 15$ (2); quotient $= 216$ (1); cube root $= 6$ (1).	1+2+1+1

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